SKYCAB - A4 PRINT PACK

1. Print at 100% / "Actual size". Turn OFF "fit to page" and "shrink oversized pages".

2. Measure the square below: it must be 10.0 cm both ways. (9.7 cm = the printer shrank it: fix the setting, or print at 103%.) Pages 2-7 must be true size; the maps can be any size.

10 cm x 10 cm

WHAT'S IN THIS PACK

p.2-5 Tube pieces for the 7 cm tube
P pod wrap, O fish-mouth (paper)
Y collar: thin card, cut 2
N cap: cardboard, cut 2
(replace the old N, O, P, Y)

p.6-7 C1 strut-hole templates
upper C1 and flipped C1, true size

p.8-12 Placement maps, sheets 1-5
for tubes running TOP to BOTTOM

REMINDERS

Tubes: turn every paper piece so its grey lines run top to bottom, along the tubes. Printed side up; never flip.

Face: the smoother face of the cardboard goes up when tracing; it becomes the outside. Same face up on all sheets.

C1 hole: cut out the template on the solid RED line, lay it on your C1 with the dashed circle and fold lines matching, trace inside. Cut the hole after the walls are glued to the deck and belt ring.

Strut: sits right up against the underside of the deck. Wire hole on the UNDERSIDE, 2 cm inside the right wall.

R3 and R8: cut each paper strip in half, trace each half + 1 cm. Wrap, then trim the ends so they BUTT together (no overlap). Card patch inside + paper tape outside on each join. R8 join at the top middle of the seat.

2 layers: A, D1, D2, E front and Q: Fevicol MR spread thin, tubes crossed, weighted flat overnight.

Q: socket layer underneath. Ignore the printed oval; draw round your real pod top.

Pods: cut with the new P. Check P and O wrap the tube with the ends just meeting.

Live page with the 3D model and the on-screen tracer: djlord7.github.io/skycab (trace.html)

SKYCAB COSTUME - CARDBOARD SKELETON STENCILS

Made for: hips 8.0" wide x 5.0" deep, waist to mid-thigh 8.0", height 42.5". Body 60 x 44 cm, rotor span 74 cm, about 860 g.

1. PRINT AT 100% / "ACTUAL SIZE". Turn OFF "fit to page" and "shrink oversized pages".

2. Measure the 10 x 10 cm square below in BOTH directions. If it is off by more than 1 mm, fix the printer setting and reprint.
3. Parts bigger than a page are split into tiles (A1, B1, A2...). Trim the margin on the RIGHT and BOTTOM edges of a tile along its grey border, lay it over the next tile so the grey borders meet and the dotted 5 cm grid lines continue. Tape.
4. Cut the paper templates out, lay them on cardboard (match the FLUTES arrow to the cardboard ridges), trace, then cut.
5. HALF patterns (red dash-dot line): trace, flip the template over along the red line, trace the other half.

10 cm x 10 cm

Measure me before cutting anything

Cut line

Fold line (glue tabs fold inward)

Press line: inside face, press with a blunt knife, do no

Centre line: flip template here

Placement mark (do not cut)

Run the cardboard flutes this way

PRINTED PARTS

N	Pod bottom cap	Cut 2 (one per pod)	Cardboard	page 4
Y	Pod top collar	Cut 2 (one per pod)	Thin card (cereal box)	page 3
O	Fish-mouth guide	Cut paper only, use 4 times	Paper (the printout itself)	page 3
P	Pod tube wrap template	Cut paper only, use for both pods	Paper (the printout itself)	page 2

RULER CUTS (measure and cut with a steel ruler, no template)

R1	Strut tube	52.0 cm long, 7 cm across, cut 1
R2	Motor pod tube	16.7 cm long (back side), 7 cm across, cut 2
R3	Rotor rim strip	6 x 71.1 cm (3 cm rim + 1.5 cm tabs each side), cut 2
R5	Wall joiner strips	5 x 9 cm (4) and 5 x 14 cm (4), cut 8
R8	Seat edge strip	7 x 64 cm (4 cm + 1.5 cm tabs each side), cut 1
R6	Shoulder straps	2.5 cm elastic, 2 x about 55 cm

10 cm check: measure this. Must be exactly 10 cm.

BACK

FRONT

strut joins here (inner side)

wire

cut: bottom of pod

P POD TUBE

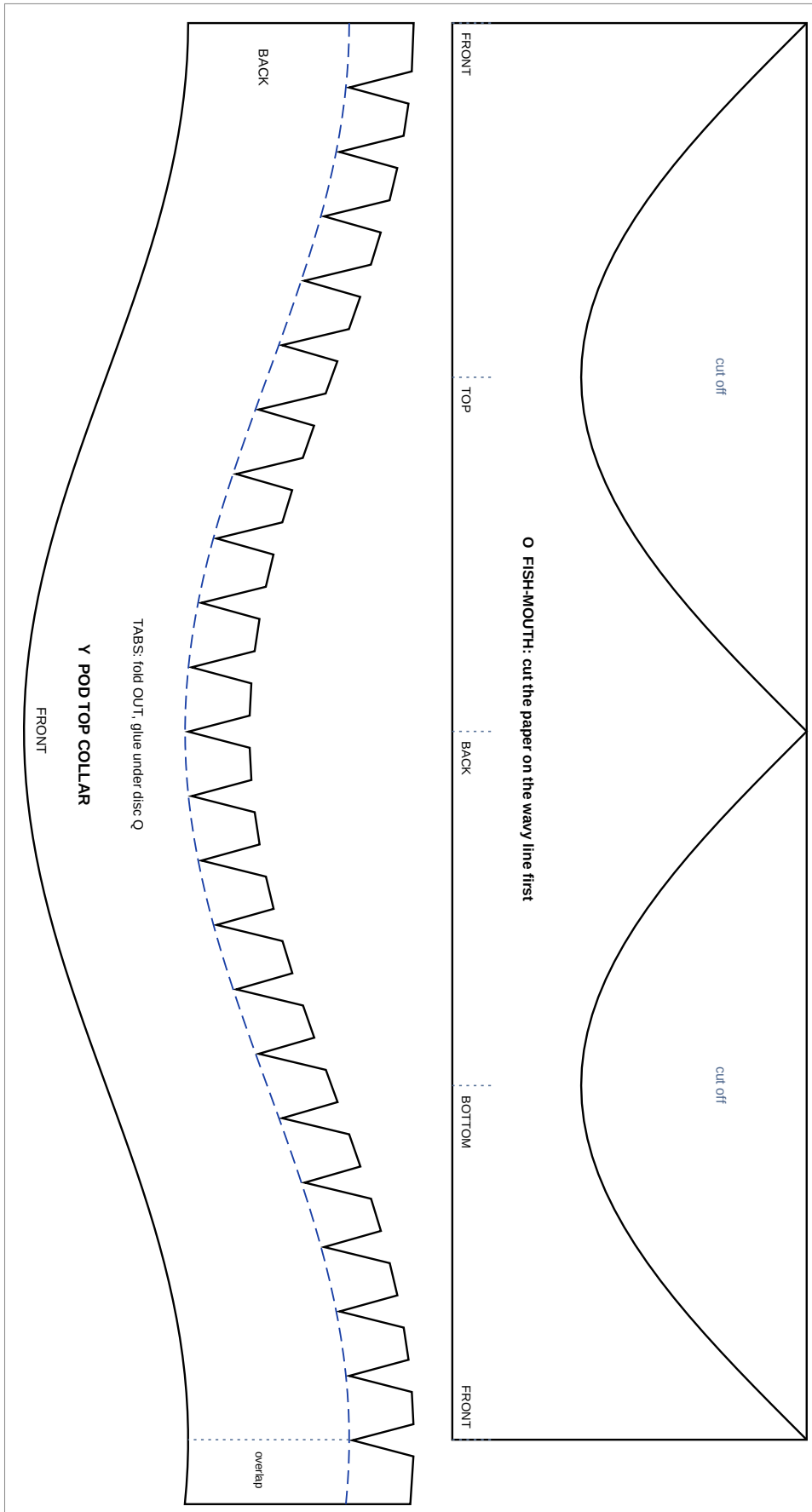
wrap round tube, top edge flush

ends must just meet: made for a

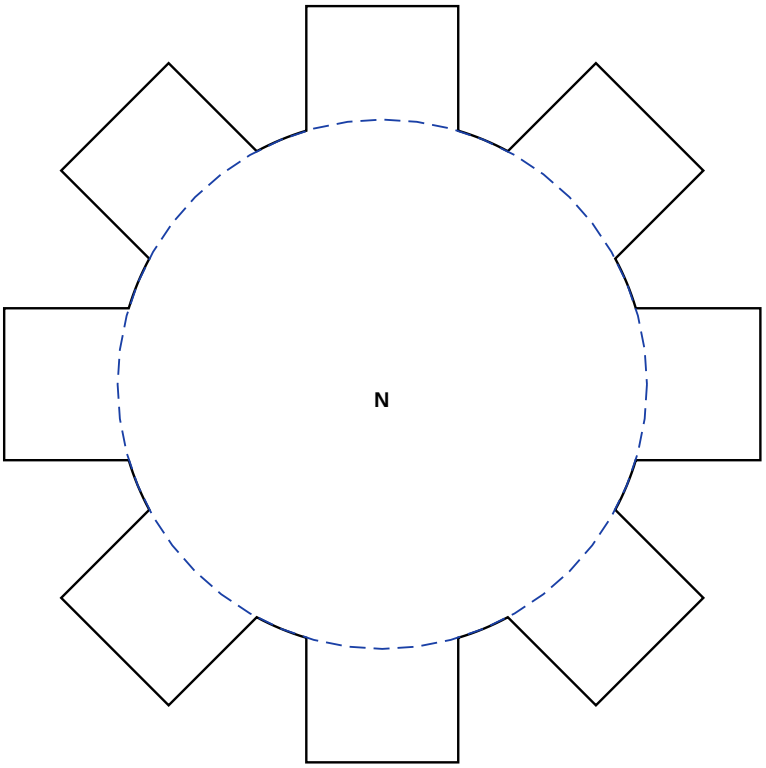
7 cm OUTSIDE diameter tube

gap or overlap? re-measure and enter it on the page

10 cm check: measure this. Must be exactly 10 cm.



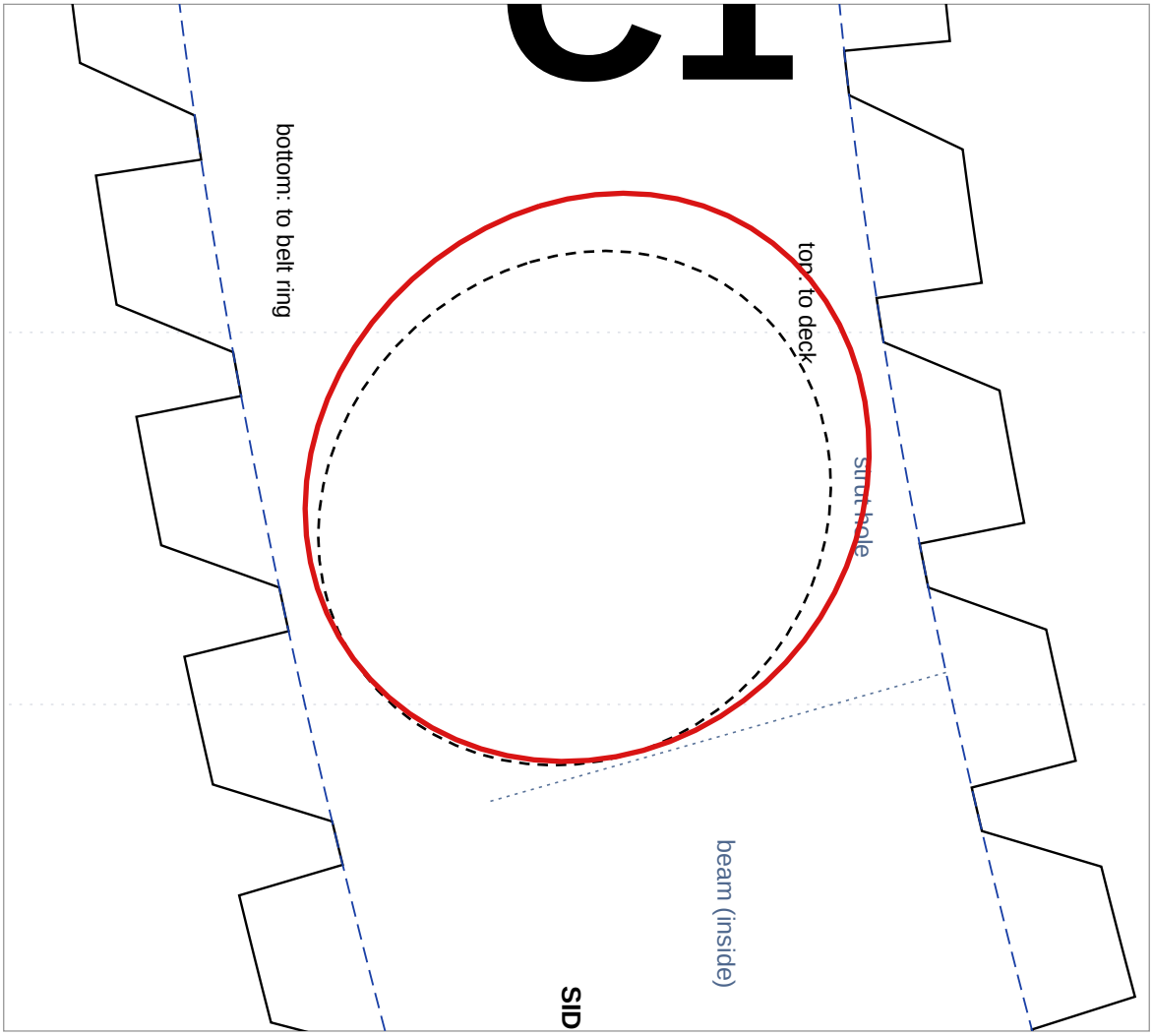
10 cm check: measure this. Must be exactly 10 cm.



TRUE SIZE - check the 10 cm ruler at the bottom before cutting.

Cut out on the solid red line (making a hole in the paper), lay the paper on your C1 so the dashed circle, blue fold lines and tab edges match your print, then trace inside the hole.

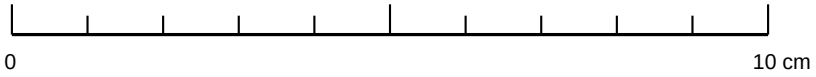
C1 strut hole, page 1 of your roll print: upper C1



dashed black = hole as printed (6.35 cm tube)

SOLID RED = new hole to cut (7 cm tube). Trace the red line.

Match the dashed circle, blue fold lines and tab edges to your print, then trace.

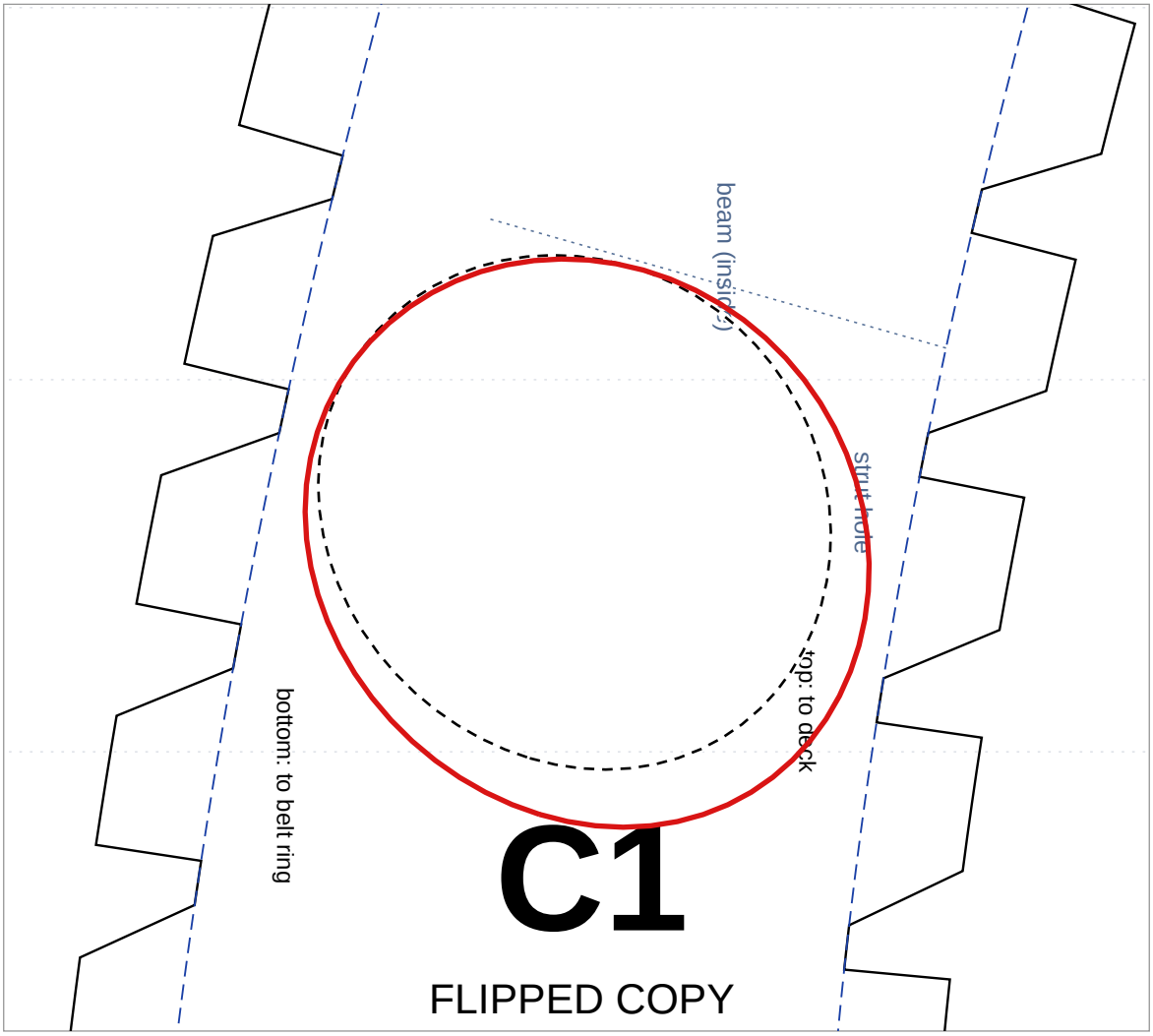


Zoom until this ruler is exactly 10 cm on a real ruler (on paper prints: check it first).

TRUE SIZE - check the 10 cm ruler at the bottom before cutting.

Cut out on the solid red line (making a hole in the paper), lay the paper on your C1 so the dashed circle, blue fold lines and tab edges match your print, then trace inside the hole.

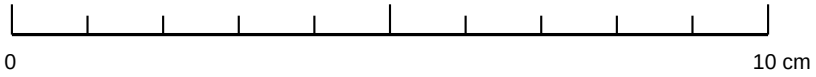
C1 strut hole, page 1 of your roll print: lower C1 (FLIPPED COPY)



dashed black = hole as printed (6.35 cm tube)

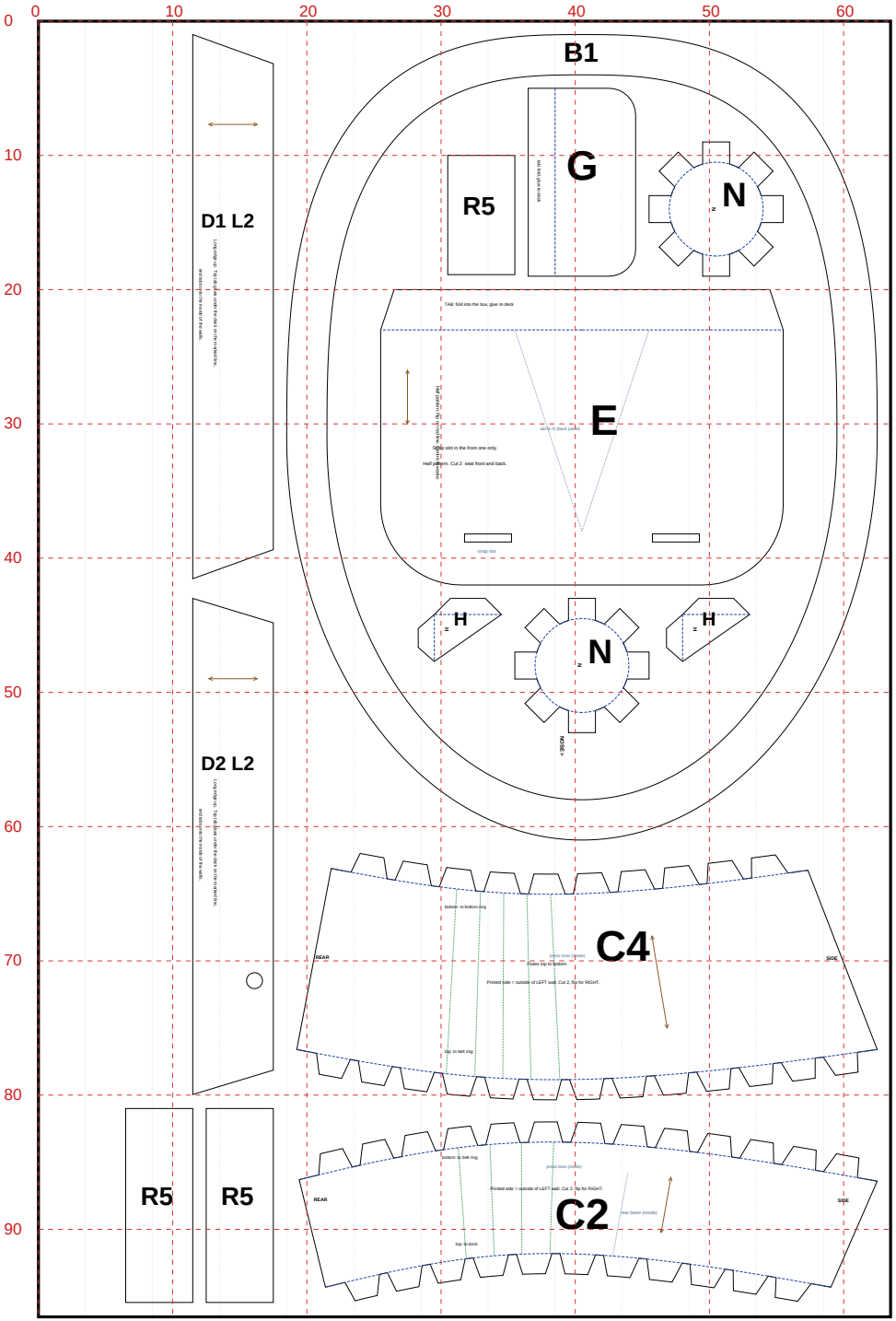
SOLID RED = new hole to cut (7 cm tube). Trace the red line.

Match the dashed circle, blue fold lines and tab edges to your print, then trace.



Zoom until this ruler is exactly 10 cm on a real ruler (on paper prints: check it first).

SHEET 1 OF 5 - for waves running UP-DOWN



Lay the cardboard UPRIGHT, like this map:

25 in (63.5 cm) edge along the top, 38 in (96.5 cm) edge down the side.

Waves inside the cardboard run UP-DOWN (top edge shows the wavy profile).

Turn each paper piece until its grey lines run UP-DOWN, then match the map.

Red numbers = cm from the TOP-LEFT corner of the cardboard.

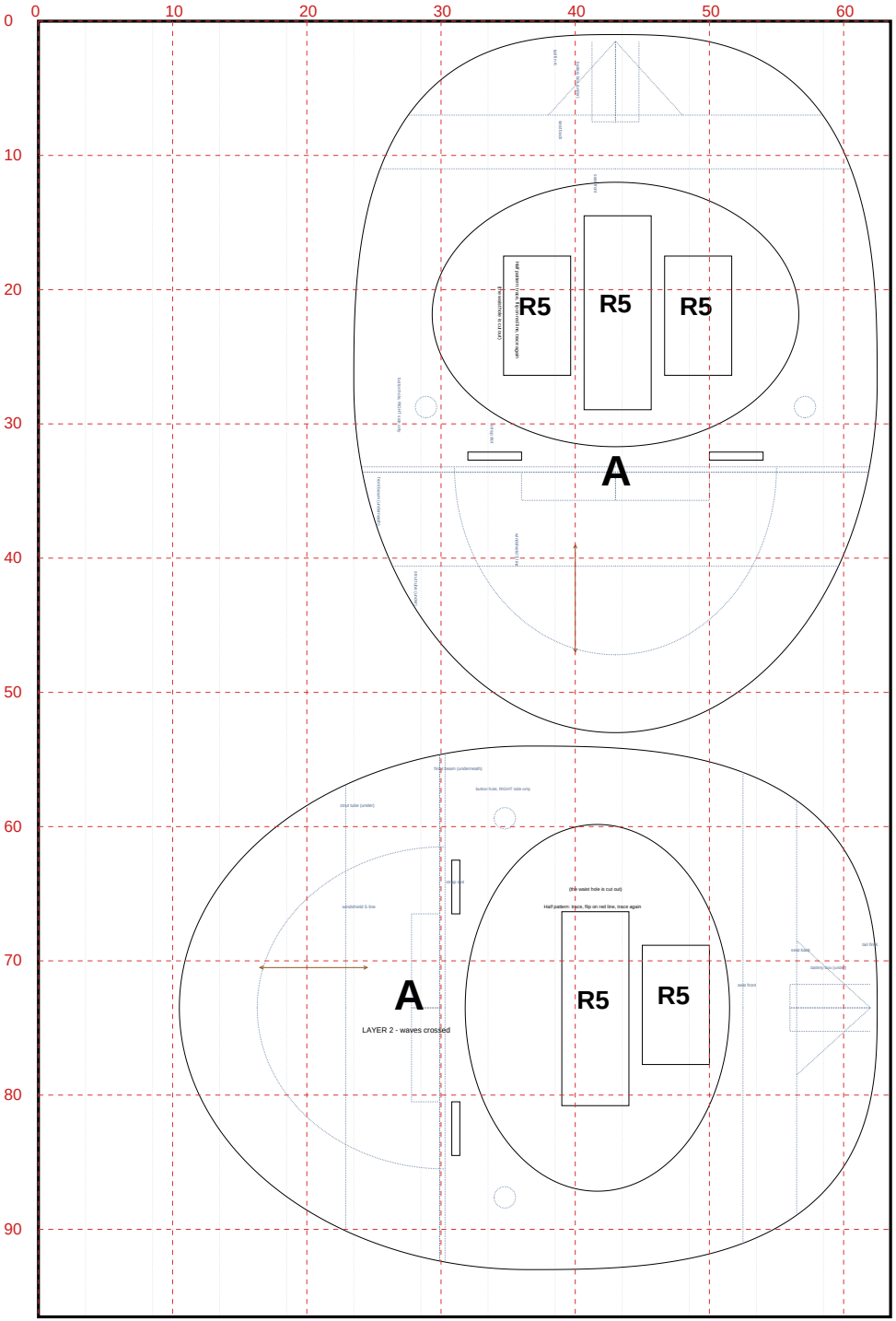
Piece	left edge	top edge	size (across x down)
D1 Front beam	**12 cm**	**1 cm**	6 x 40 cm
B1 Belt ring	**18 cm**	**1 cm**	44 x 60 cm
G Battery box parts	**36 cm**	**5 cm**	8 x 14 cm
N Pod bottom cap (use NEW A4 one)	**46 cm**	**9 cm**	10 x 10 cm
R5 Wall joiner strip	**30 cm**	**10 cm**	5 x 9 cm
E Seat	**26 cm**	**20 cm**	30 x 22 cm
D2 Rear beam	**12 cm**	**43 cm**	6 x 37 cm
H Headlight part	**28 cm**	**43 cm**	6 x 5 cm
N Pod bottom cap (use NEW A4 one)	**36 cm**	**43 cm**	10 x 10 cm
H Headlight part	**47 cm**	**43 cm**	6 x 5 cm
C4 Wall lower rear	**19 cm**	**62 cm**	43 x 18 cm
R5 Wall joiner strip	**6 cm**	**81 cm**	5 x 14 cm
R5 Wall joiner strip	**12 cm**	**81 cm**	5 x 14 cm
C2 Wall upper rear	**19 cm**	**82 cm**	43 x 13 cm

"left edge" = how far the piece's left-most point sits from the cardboard's left edge; "top edge" = how far its top-most point is from the top.

Order: place the BIG pieces first, then fit small ones in the gaps and inside the rings (they come out of the ring's middle - cut those out carefully).

N: use the new N from the A4 file in this spot.

SHEET 2 OF 5 - for waves running UP-DOWN



Lay the cardboard UPRIGHT, like this map:

25 in (63.5 cm) edge along the top, 38 in (96.5 cm) edge down the side.

Waves inside the cardboard run UP-DOWN (top edge shows the wavy profile).

Turn each paper piece until its grey lines run UP-DOWN, then match the map.

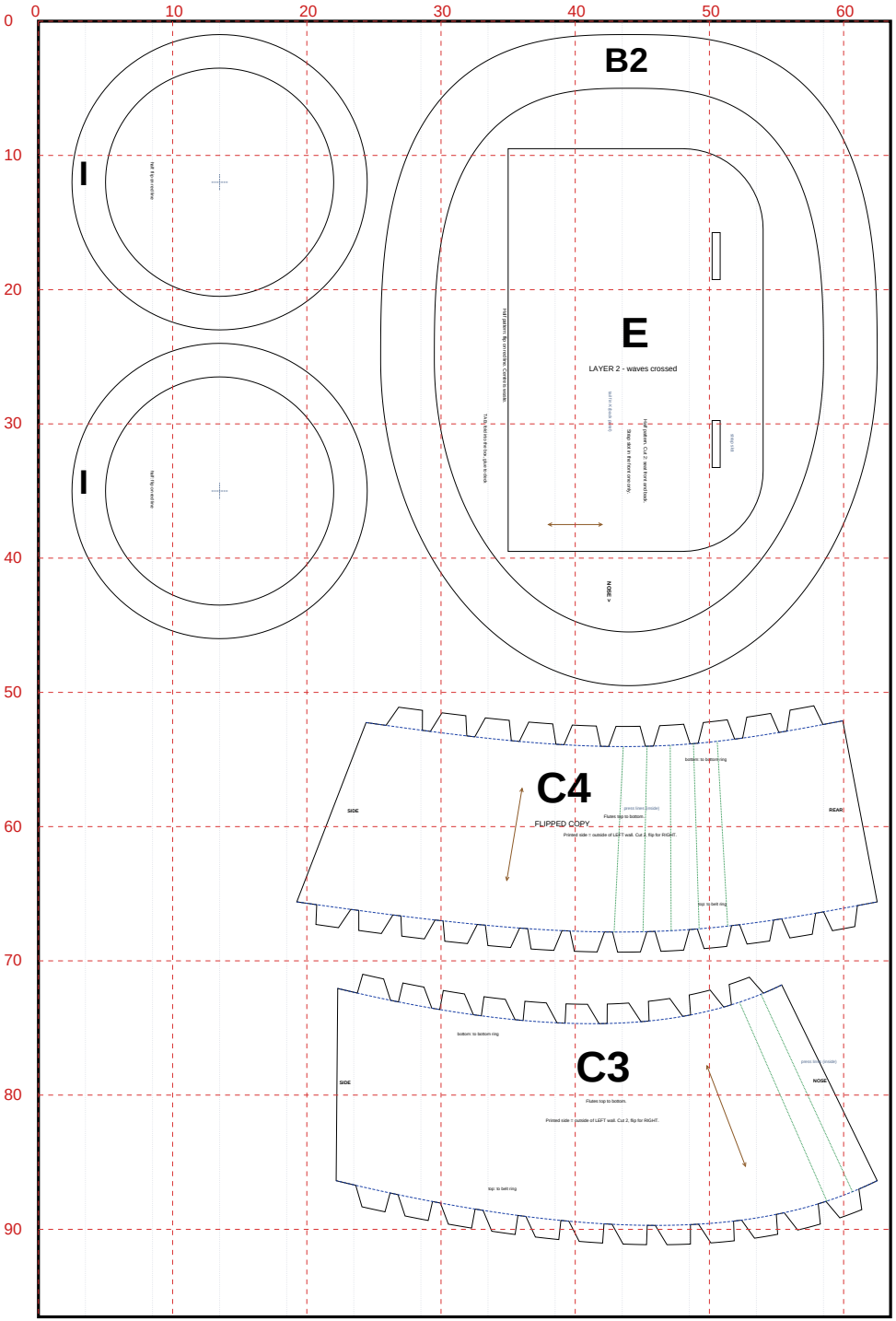
Red numbers = cm from the TOP-LEFT corner of the cardboard.

Piece	left edge	top edge	size (across x down)
A Deck	**24 cm**	**1 cm**	39 x 52 cm
R5 Wall joiner strip	**41 cm**	**14 cm**	5 x 14 cm
R5 Wall joiner strip	**35 cm**	**18 cm**	5 x 9 cm
R5 Wall joiner strip	**47 cm**	**18 cm**	5 x 9 cm
A Deck	**10 cm**	**54 cm**	52 x 39 cm
R5 Wall joiner strip	**39 cm**	**66 cm**	5 x 14 cm
R5 Wall joiner strip	**45 cm**	**69 cm**	5 x 9 cm

"left edge" = how far the piece's left-most point sits from the cardboard's left edge; "top edge" = how far its top-most point is from the top.

Order: place the BIG pieces first, then fit small ones in the gaps and inside the rings (they come out of the ring's middle - cut those out carefully).

SHEET 3 OF 5 - for waves running UP-DOWN



Lay the cardboard UPRIGHT, like this map:

25 in (63.5 cm) edge along the top, 38 in (96.5 cm) edge down the side.

Waves inside the cardboard run UP-DOWN (top edge shows the wavy profile).

Turn each paper piece until its grey lines run UP-DOWN, then match the map.

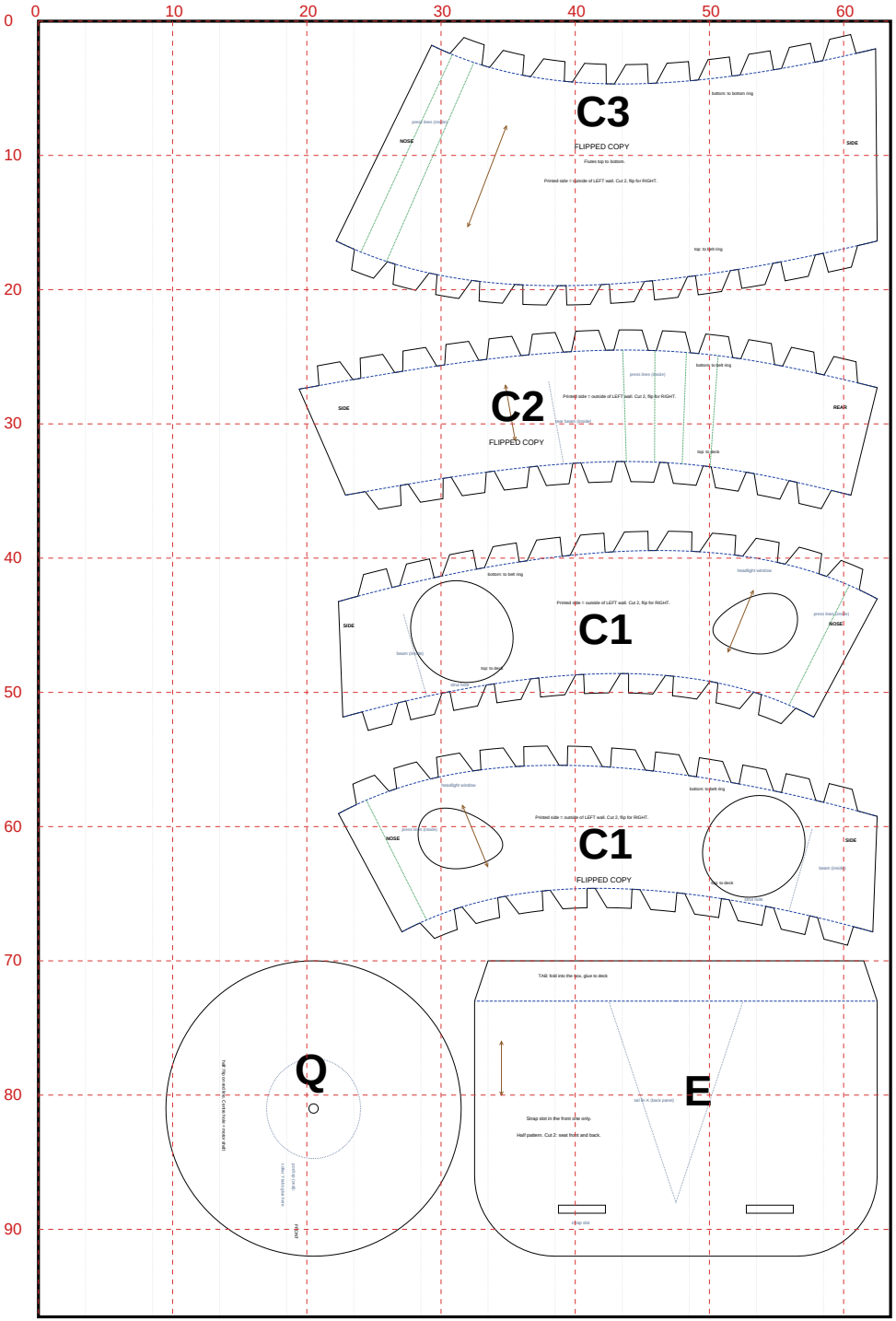
Red numbers = cm from the TOP-LEFT corner of the cardboard.

Piece	left edge	top edge	size (across x down)
I Rotor lip ring	**2 cm**	**1 cm**	22 x 22 cm
B2 Bottom ring	**26 cm**	**1 cm**	37 x 48 cm
E Seat	**35 cm**	**10 cm**	19 x 30 cm
I Rotor lip ring	**2 cm**	**24 cm**	22 x 22 cm
C4 flipped copy Wall lower rear	**19 cm**	**51 cm**	43 x 18 cm
C3 Wall lower front	**22 cm**	**71 cm**	40 x 20 cm

"left edge" = how far the piece's left-most point sits from the cardboard's left edge; "top edge" = how far its top-most point is from the top.

Order: place the BIG pieces first, then fit small ones in the gaps and inside the rings (they come out of the ring's middle - cut those out carefully).

SHEET 4 OF 5 - for waves running UP-DOWN



Lay the cardboard UPRIGHT, like this map:

25 in (63.5 cm) edge along the top, 38 in (96.5 cm) edge down the side.

Waves inside the cardboard run UP-DOWN (top edge shows the wavy profile).

Turn each paper piece until its grey lines run UP-DOWN, then match the map.

Red numbers = cm from the TOP-LEFT corner of the cardboard.

Piece	left edge	top edge	size (across x down)
C3 flipped copy Wall lower front	**22 cm**	**1 cm**	40 x 20 cm
C2 flipped copy Wall upper rear	**19 cm**	**23 cm**	43 x 13 cm
C1 Wall upper front	**22 cm**	**38 cm**	40 x 15 cm
C1 flipped copy Wall upper front	**22 cm**	**54 cm**	40 x 15 cm
Q Rotor base disc	**10 cm**	**70 cm**	22 x 22 cm
E Seat	**32 cm**	**70 cm**	30 x 22 cm

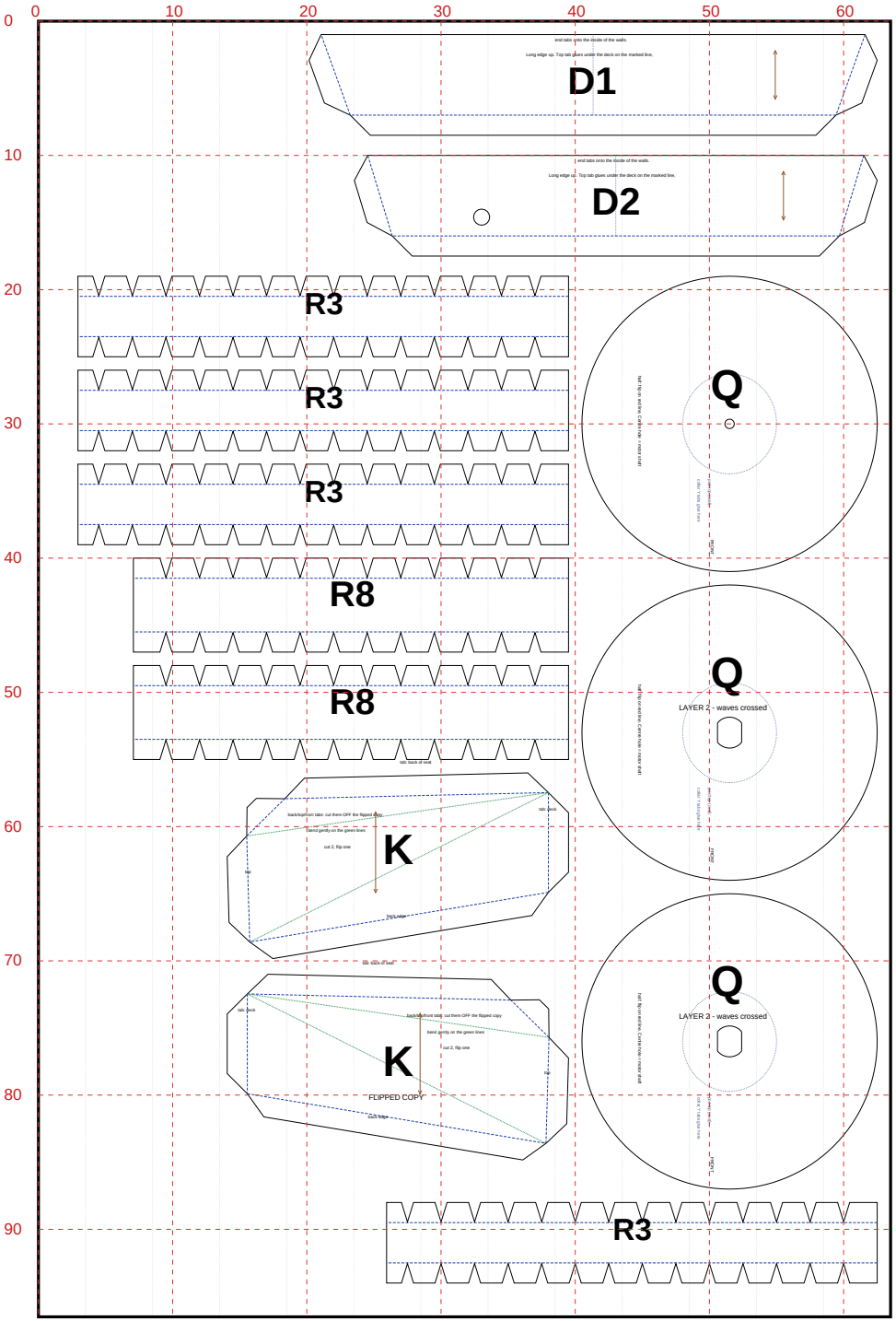
"left edge" = how far the piece's left-most point sits from the cardboard's left edge; "top edge" = how far its top-most point is from the top.

Order: place the BIG pieces first, then fit small ones in the gaps and inside the rings (they come out of the ring's middle - cut those out carefully).

C1: the map shows the new strut hole; cut it after gluing.

Q: ignore the printed oval; draw round your real pod top.

SHEET 5 OF 5 - for waves running UP-DOWN



Lay the cardboard UPRIGHT, like this map:

25 in (63.5 cm) edge along the top, 38 in (96.5 cm) edge down the side.

Waves inside the cardboard run UP-DOWN (top edge shows the wavy profile).

Turn each paper piece until its grey lines run UP-DOWN, then match the map.

Red numbers = cm from the TOP-LEFT corner of the cardboard.

Piece	left edge	top edge	size (across x down)
D1 Front beam	**20 cm**	**1 cm**	42 x 8 cm
D2 Rear beam	**24 cm**	**10 cm**	39 x 8 cm
R3 Rotor rim strip	**3 cm**	**19 cm**	37 x 6 cm
Q Rotor base disc	**40 cm**	**19 cm**	22 x 22 cm
R3 Rotor rim strip	**3 cm**	**26 cm**	37 x 6 cm
R3 Rotor rim strip	**3 cm**	**33 cm**	37 x 6 cm
R8 Seat edge strip	**7 cm**	**40 cm**	32 x 7 cm
Q Rotor base disc	**40 cm**	**42 cm**	22 x 22 cm
R8 Seat edge strip	**7 cm**	**48 cm**	32 x 7 cm
K Tail fin	**14 cm**	**56 cm**	25 x 14 cm
Q Rotor base disc	**40 cm**	**65 cm**	22 x 22 cm
K flipped copy Tail fin	**14 cm**	**71 cm**	25 x 14 cm
R3 Rotor rim strip	**26 cm**	**88 cm**	37 x 6 cm

"left edge" = how far the piece's left-most point sits from the cardboard's left edge; "top edge" = how far its top-most point is from the top.

Order: place the BIG pieces first, then fit small ones in the gaps and inside the rings (they come out of the ring's middle - cut those out carefully).

R3 / R8 strips are too long for this way round, so each is made in 2 halves: cut your paper strip across the middle, trace each half and add 1 cm at the cut end. Wrap, trim so the ends BUTT together (no overlap). R3: 4 halves; R8: 2.

Q: ignore the printed oval; draw round your real pod top.